

Sama Mushtaq

Mechanical Design and Testing

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Engineering Experience

MKEC Engineering - Mechanical Design Engineer

December 2019 to December 2020

- Prepared heating and cooling load calculations and mechanical documentation for 50+ commercial, residential and modular building projects using AutoCAD, Revit and Trane TRACE.
- Produced duct layouts, HVAC elevations, equipment schedules and diffuser schedules within the team's design, checking and revision process.
- Used BIM 360 and coordinated design information with architects, engineers, project managers and contractors. Developed reusable Revit families, templates and details.

Selected Engineering Projects

Vibration Sensor Response - Wichita State University

2016

- Helped assemble a two sensor vibration setup as part of a team. Analyzed the recorded response and sensitivity differences across three LabVIEW channels and a preserved run of 1,022 samples.

Kite Buggy Structural Analysis - Wichita State University

2015

- Shared load, stress and deflection calculations with two teammates on a five-person design team working to weight and cost limits. Another teammate produced the CATIA model.

Shell-and-Tube Heat Exchanger Study - Wichita State University

2016

- Completed an individual calculation study of a countercurrent water-to-oil heat exchanger, examining heat duty, heat-transfer coefficients, fouling, material properties and dimensions.

Additional Experience

Mercor - AI Quality Reviewer, Project-Based Independent Contractor

February 2026 to Present

- Review technical and business outputs against source records, calculations and project requirements. Document unsupported claims, numerical errors and missing requirements.

RYGNeco - Co-Founder and Program Lead

August 2024 to Present

- Developed procedures and product requirements for electronics intake, testing, custody and disposition, supporting a pilot covering approximately 130 devices.

Hard Rock Casino Cincinnati - Operations Management

January 2024 to January 2025

- Advanced through operations manager, assistant general manager and acting general manager responsibilities across three outlets and approximately 120 staff. Led staffing, training and inventory controls.

Education

University of Cincinnati - Mechanical Engineering

Admitted for Spring 2027. Transfer evaluation and completion schedule in progress.

Wichita State University - Mechanical Engineering Studies

2012 to 2016

103 earned credits. GPA 3.397. Dean's List in Fall 2012, Spring 2014 and Fall 2015.

Technical Skills and Coursework

AutoCAD, Autodesk Revit, Trane TRACE, BIM 360, Microsoft Excel, LabVIEW data analysis, Python, HVAC documentation, structural calculations and heat transfer. ME 637 CNC coursework and hands-on laboratory work, including G-code.